3.5 Preparation of Materials for Staining

The essential steps in the preparation of materials to be observed are

1) Preparation of smear
2) Fixation
3) Application of one or more staining solutions

3.5.1 Preparation of Smear

Smears can be made from liquid or solid cultures or from clinical specimens. Smear is prepared by placing a loopful of culture on a clear glass slide with an inoculation loop. The culture is spread on the glass slide so as to form a thin film. This film is allowed to air dry (Figure 3.4).

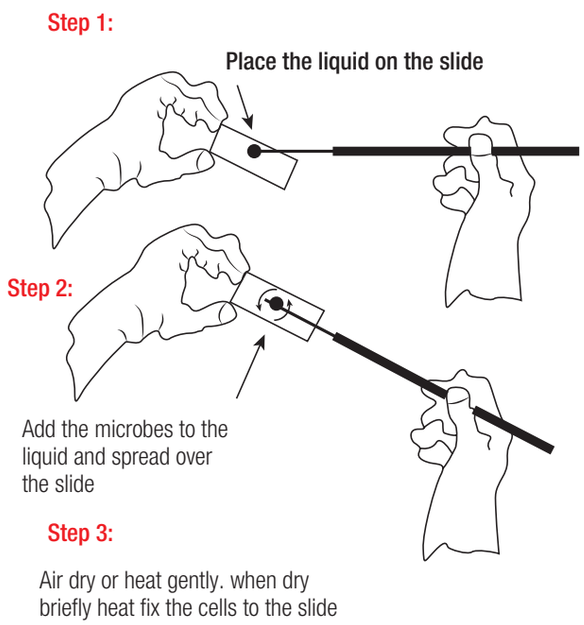


Figure 3.4: Preparation of smear

3.5.2 Fixation

Fixation kills the microorganisms and attaches them to the slide. This prevents washing away of microorganism in further steps of staining procedure. It also preserves various parts of microorganisms in their natural state with only minimal distortion. The two fixation methods that are used to fix microbial cells are heat fixation and chemical fixation.

Heat fixation

In this method the slide is gently heated by passed through a flame (Figure 3.5). Heat fixation will preserve the overall morphology of the cell without destroying the internal structures.

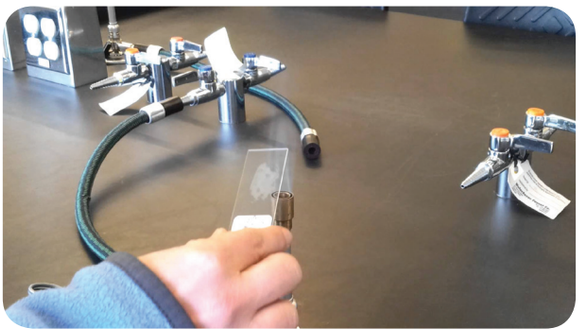


Figure 3.5: Fixation of smear by passing slide gently through the flame

Chemical fixation

It involves the use of chemical fixative to protect the fine cellular structures of delicate microorganisms. For this purpose, Ethanol, Acetic acid, Formaldehyde, Glutaraldehyde and Mercuric chloride are usually used.

3.5.3 Bacterial Staining Methods

Different staining methods are employed to study the bacterial morphology and to identify bacteria. Some methods are used for general purposes and others are used for special purposes. There are three categories of staining methods, they are:

DO YOU KNOW?

Robert Hooke was the first to describe the appearance of stained objects under light microscope.

Professor Joseph Von Gerlach of Germany was the first to use stain in histology.

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